

Elijah Ifechukwu Nelson

Machine Learning & Embedded Systems Engineer

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Summary

Machine Learning and Embedded Systems Engineer completing an M.Phil. in Machine Learning and Machine Intelligence at the University of Cambridge, after graduating as Covenant University's valedictorian in Electrical & Electronics Engineering. Experienced in building end-to-end systems spanning low-power firmware, sensor platforms, robotics, and production ML infrastructure. Delivered 1,000+ field-deployed climate instruments across four African countries and co-authored published research on tactile slip classification for robotic manipulation. Interested in efficient, reliable ML systems for resource-constrained and real-world deployment. **Education**

M.Phil. in Machine Learning and Machine Intelligence, University of Cambridge, Cambridge, UK 10/2025 - 08/2026

- Advanced coursework in Probabilistic Machine Learning, Optimization, and Reinforcement Learning.
- Thesis supervised by Andrew Fitzgibbon: investigating whether structured Kronecker-factored and low-rank approximations of transformer Hessians can match or outperform Muon, Shampoo, and SOAP during language-model pretraining. Implementing Muon and Shampoo from scratch, analysing full-Hessian and Gauss–Newton spectra on small transformers, and evaluating scaling behaviour under the Wen et al. (2025) fair-comparison protocol.

B.Eng. in Electrical and Electronics Engineering, Covenant University, Ota, Nigeria 08/2018 - 09/2023

- First Class Honours with a 4.98/5.0 CGPA, the highest in the department's history; ranked 1st of 132 in the programme and 1st of 1,175 university-wide.
- Designed, simulated, and built a multipurpose autonomous UAV featuring autonomous navigation, precision landing, and visual target tracking. [link]

Research Interests

Machine Learning Systems, Efficient Optimization, Embedded AI, Robotics, Sensor Intelligence, Computer Architecture, Embedded Software and Operating Systems, and Resource-Constrained Deployment.

Research & Teaching Experience

AI Research Assistant, Intelligent Manipulation Lab, University of Lincoln, Lincoln, UK 03/2022 - 06/2023

- Developed a novel SMOTE-CNN pipeline for real-time slip classification from tactile data without kinematic-state inputs, achieving 98% accuracy and reducing false negatives by 95.43% on a limited, highly imbalanced dataset. [link]
- Benchmarked DNN, RNN, CNN, LSTM, ConvLSTM, and Vision Transformer architectures across 1-D, 2-D, 3-D, and 4-D tactile representations to quantify trade-offs in detection accuracy, prediction horizon, and inference latency.
- Evaluated SHAP and saliency-based explainability approaches, then applied Grad-CAM to 16×16 uSkin tactile images across 16 representative test trials, showing that edge and central taxels with pronounced shear-force changes dominated SMOTE-CNN slip decisions.

Research Assistant, Dept. of Electrical & Electronics Engineering, Covenant University, Ota, Nigeria 01/2023 - 07/2023

- Developed ATmega328-PU firmware for an automated waste-management unit and an autonomous vacuum-cleaning robot, integrating sensing, actuation, communications, and control.
- Implemented interrupt-driven ultrasonic sensing, low-power scheduling, and SIM800L-based SMS alerts for the waste unit, delivering reliable operation across all tested configurations.

- Built a ROS-based navigation stack for the vacuum robot, with a laptop master node executing navigation and computer-vision algorithms, exchanging odometry and motion commands through ROS nodes; validated the system in Gazebo and via game-controller teleoperation.
- Trained 50+ students in embedded-systems research methods, experimental design, and prototyping.

Tutor, Covenant University, Ota, Nigeria

01/2021 - 07/2023

- Taught 200+ students in Electromagnetism, Applied Electronics, Advanced Control Systems, Power Electronics, and High Voltage Engineering using original notes, worked examples, and applied case studies.
- Achieved an average pass rate of 85% across supported courses, including 77% in Advanced Control Systems.

Publications

- N. Elijah, K. Nazari, M. Esfandiari, and A. Ghalamzan-E, “Advancing Slip Classification in Robotic Manipulation Through Tactile Data Representation and Model Selection,” *IEEE Sensors Journal*, vol. 25, no. 17, 2025. [link]
- T. Somefun, N. Elijah, A. E. Igho-Orere, O. Timilehin, C. Somefun, and S. Ongbali, “Energy Optimization Algorithm for Reducing Energy Consumption in a Smart Home,” *2023 2nd International Conference on Multidisciplinary Engineering and Applied Science (ICMEAS)*, pp. 1–5, 2023. [link]

Professional Experience

Founding Systems Engineer, Climate in Africa, Lagos, Nigeria

10/2023 - 08/2025

- Engineered low-cost, WMO-compliant Modular In-situ Reporting Instruments (MIRI) measuring 14 meteorological parameters; supported deployment of 1,000+ units across Nigeria, Ivory Coast, Kenya, and Ghana.
- Developed a NOR Flash driver for the MT25QL01GBBB and co-designed CRC-verified over-the-air firmware updates over BLE and octet streams, enabling resilient field upgrades; ported approximately 3,000 lines of firmware from ATmega to STM32 F4/F103, improving performance by 30%.
- Extended MIRI battery life to 3.5 days by implementing an RTOS-based sleep scheduler and non-blocking state machines; reduced GSM/GPRS communication latency by approximately 70% and added automatic NOR Flash/SD-card fallback logging during connectivity loss.
- Architected the email and notification infrastructure for climateinafrica.com, which aggregates terabytes of monitor and user data, and built a NOAA ingestion pipeline to scrape, parse, and expose gigabyte-scale climate datasets within the analytics sandbox.
- Owned end-to-end technical and product documentation for the platform, translating system architecture, firmware behaviour, deployment procedures, and operational workflows into maintainable engineering references.
- Developed an autonomous low-Earth-orbit satellite tracker using non-blocking task scheduling for azimuth/elevation actuation and a custom LoRa ground-station link for telemetry and control.
- Validated UHF/VHF satellite radios using spectrum and logic analysers, diagnosing signal-integrity and communications issues before field deployment.

Machine Learning Engineer, Ticki, Remote

08/2024 - 07/2025

- Architected a facial-similarity retrieval service using OpenCV, FAISS, Celery, and Redis for 5,000 users; reduced end-to-end latency by approximately 70% to below 100 ms by migrating from IndexFlatL2 to IndexIVFFlat and caching embeddings in memory.
- Built the complete indexing and retrieval pipeline: detected faces across the source image repository, generated traceable NumPy embeddings keyed to filenames, and matched query-face embeddings against the FAISS index.
- Supported multi-face queries through an interactive selection workflow using in-memory BytesIO buffers; distributed face detection and embedding generation to Celery workers and used Redis to deduplicate previously processed images across an approximately 3,000-image dataset.
- Hardened and scaled the Ubuntu VPS deployment for near-zero downtime under 1,000+ concurrent requests using IP-restricted UFW rules, least-privilege user groups, multiple Gunicorn workers, and Nginx load balancing.

Reservoir Performance Simulation Intern, Schlumberger (SLB), Port
Harcourt, Nigeria

04/2022 - 08/2022

- Characterised drilling and formation-stability parameters through wet-lab rock and fluid analysis, including rheology, density/PVT, calorimetry, and precision measurements; interpreted subsurface and wellbore data in Petrel and Techlog.
- Inspected, serviced, and validated CPS-361 surface units, compressors, generators, and associated flow, pressure, temperature, and optical sensors against pre-deployment checklists, increasing maintenance throughput by approximately 15%; also proposed an automated CPS-361 design concept.
- Managed IT operations across the SLB base, resolving hardware, software, printer, router, access-point, and user-provisioning issues while administering the support-ticket queue.

Delegate of Croatia, Model United Nations, Beirut, Lebanon

03/2019 - 04/2019

- Researched the protection of Indigenous cultural heritage and cultural artefacts, producing policy briefs and committee reports for UNESCO deliberations.

Delegate of Israel, Model United Nations, Buenos Aires, Argentina

09/2018 - 10/2018

- Researched and debated policy mechanisms for mitigating climate change impacts in developing economies.

Selected Projects

CamAI: Action-Recognition Camera System, MediaPipe, ConvLSTM,
Computer Vision

03/2023 - 05/2023

- Led a two-person team in developing a patient-monitoring camera system that detects and classifies human actions from live video.
- Extracted pose landmarks with MediaPipe and trained a ConvLSTM ensemble that classified patient states with 96%+ accuracy, maintained reliable subject tracking, and streamed results to a web interface.

Quad-X: Quadcopter Flight Controller, ATmega328-PU, PID, Sensor Fusion

01/2021 - 12/2021

- Designed and built an ATmega328-PU-based quadcopter flight controller, fusing IMU measurements within PID control loops to achieve stable attitude control across all axes.
- Extended the platform into AutoUAV, adding autonomous navigation, in-flight object detection, precision landing, and live camera streaming.

MilliMeter: Low-Cost IoT Metering System, Embedded Communications,
Web Technologies

03/2020 - 07/2020

- Led a four-person team in designing a low-cost IoT smart-metering platform with custom embedded communication protocols and a web-based monitoring interface for the Lagos State Smart Meter Hackathon.

Skills

- **Programming:** Python, C, C++, JavaScript, SQL.
- **Machine Learning & Computer Vision:** PyTorch, TensorFlow, scikit-learn, NumPy, FAISS, OpenCV, MediaPipe, Matplotlib, seaborn; CNNs, RNNs/LSTMs, ConvLSTMs, Vision Transformers, model interpretability.
- **Backend, Deployment & Infrastructure:** Flask, Celery, Redis, Gunicorn, Nginx, Linux/Ubuntu, UFW, Git, GitHub.
- **Embedded Systems & Communications:** STM32, ATmega, RTOS, I2C, SPI, UART, BLE, LoRa, GSM/GPRS, sensor integration, soldering, KiCad.
- **Robotics & Simulation:** ROS, Gazebo, PID control, sensor fusion, autonomous navigation.
- **Professional:** Technical writing, teaching, project leadership, stakeholder communication, experimental design, and systems thinking.

Training & Certifications

- **Deep Learning Specialization**, DeepLearning.AI (Coursera).
- **Computer Architecture Essentials on Arm**, Arm EducationX.

Leadership Experience

Administrative Head, Luminous Life Foundation, Lagos, Nigeria 09/2023 - Present

- Led a 15-person team delivering solar electricity to 100 homes, benefiting approximately 400 residents in rural Nigerian communities that had lacked reliable power for more than 50 years.

Academic Officer, Dept. of Electrical and Electronics Engineering, Covenant University, Ota, Nigeria 09/2022 - 08/2023

- Represented 500+ students as the department's principal academic liaison to 20 faculty members, resolving academic concerns and coordinating communication across cohorts.
- Organised peer tutorials, maintained past-question repositories with Tesla's Academic Group, and delivered industry-facing symposiums for Electrical Engineering students.

Volunteer Work & Community Service

Co-founder, Tech for Progress, Isolo, Lagos, Nigeria 01/2023 - Present

- Launched Isolo's first digital-skills programme for underserved youth, training 50+ secondary-school students in Microsoft Word, PowerPoint, Excel, and Canva.
- Expanded the programme toward a train-the-trainer model for teachers, supporting continuity and helping participants progress into further technology bootcamps.

Career Advisor, Tesla's Academic Group (TAG), Covenant University, Ota, Nigeria 03/2022 - 03/2023

- Supported the academic and career development of 400+ students through targeted webinars, mentoring, and practical guidance.

Lead AI Tutor, Google Developer Student Committee (GDSC), Covenant University, Ota, Nigeria 01/2022 - 12/2022

- Grew the AI learning community from 200 to 500+ members by designing and delivering a structured curriculum.
- Led hands-on natural-language-processing and computer-vision projects, guiding participants from foundational concepts to working prototypes.

Member / Assistant Project Manager (National Champions), Enactus, Covenant University, Ota, Nigeria 11/2019 - 09/2022

- Led development of ECOBAG, a heat-retention cooler designed to help small businesses reduce spoilage during unreliable power supply.
- Co-developed PET-City, a circular-economy initiative converting plastic waste into construction bricks; contributed to the team's national championship among 18 teams in 2020.

Tutor, Deeper Life Student Outreach (DLSO), Ojo, Lagos, Nigeria 07/2019 - 08/2019

- Prepared 20+ students for university entrance examinations and provided academic and career guidance.

Honors & Awards

- **Runner-Up, Fitzcelerate**, University of Cambridge – £2,500 award (2026).
- **Charitable and Community Award**, Fitzwilliam College, University of Cambridge (2026).
- **Mastercard Foundation Scholar**, University of Cambridge – selected as 1 of 69 from 4,218 applicants (2025).
- **University Valedictorian / Best Graduating Student**, Covenant University – ranked 1st of 1,175 graduates (2023).
- **Best Graduating Student, College of Engineering and EIE Department** – ranked 1st of 132; highest departmental CGPA on record, 4.98/5.0 (2023).
- **Dean's Award for Academic Excellence**, Covenant University – selected as 1 of 386 students (2023).
- **Femi Ademiju Award for Excellence** and **CUALA Best Graduating Student Award** (2023).
- **First Runner-Up**, 3-line Finance Hackathon (2022–2023) and Covenant University Innovation Hackathon

(2022).

- **Goodwall Challenge Awards:** \$100 OMGRobotics Prize and \$250 My Passion Prize (2021).
- **Winner**, Thinkerthon; **First Runner-Up**, Pan-African Hackathon; **Qualifier**, Lagos State Smart Meter Hackathon (2020).
- **Best IGCSE Student**, including top performance in Chemistry and Geography, DLHS Lagos (2018).

Media Appearances

- Best Graduating Student in Covenant University – featured (09/2023).
- 3line’s Hackathon – featured (04/2023).

Interests

Piano, cycling, running, reading, writing, blogging, coaching, and recreational coding.